



ITU Overview and WTSA-08 Results:

Forum on Next Generation Network Standardization

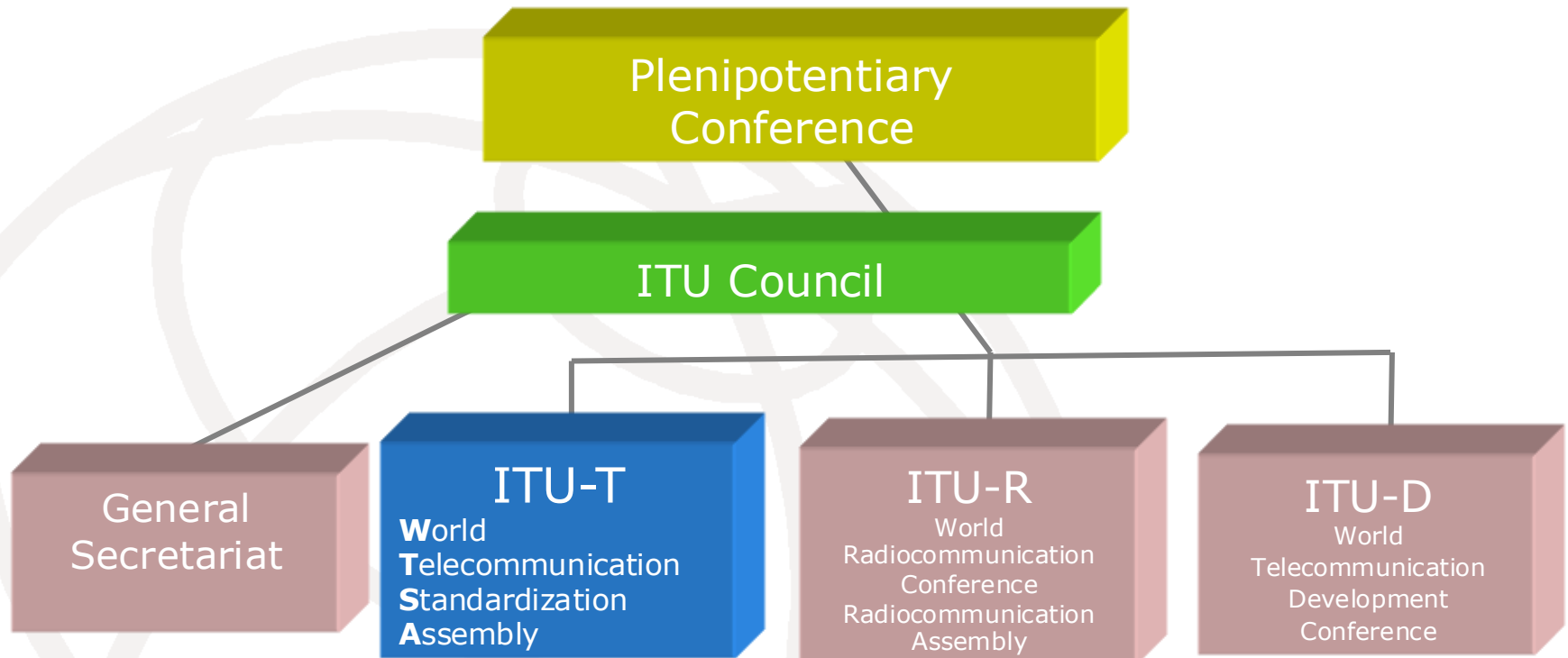
Colombo, Sri Lanka, 07-10 April 2009

Malcolm Johnson

Director, Telecommunication Standardization Bureau, ITU

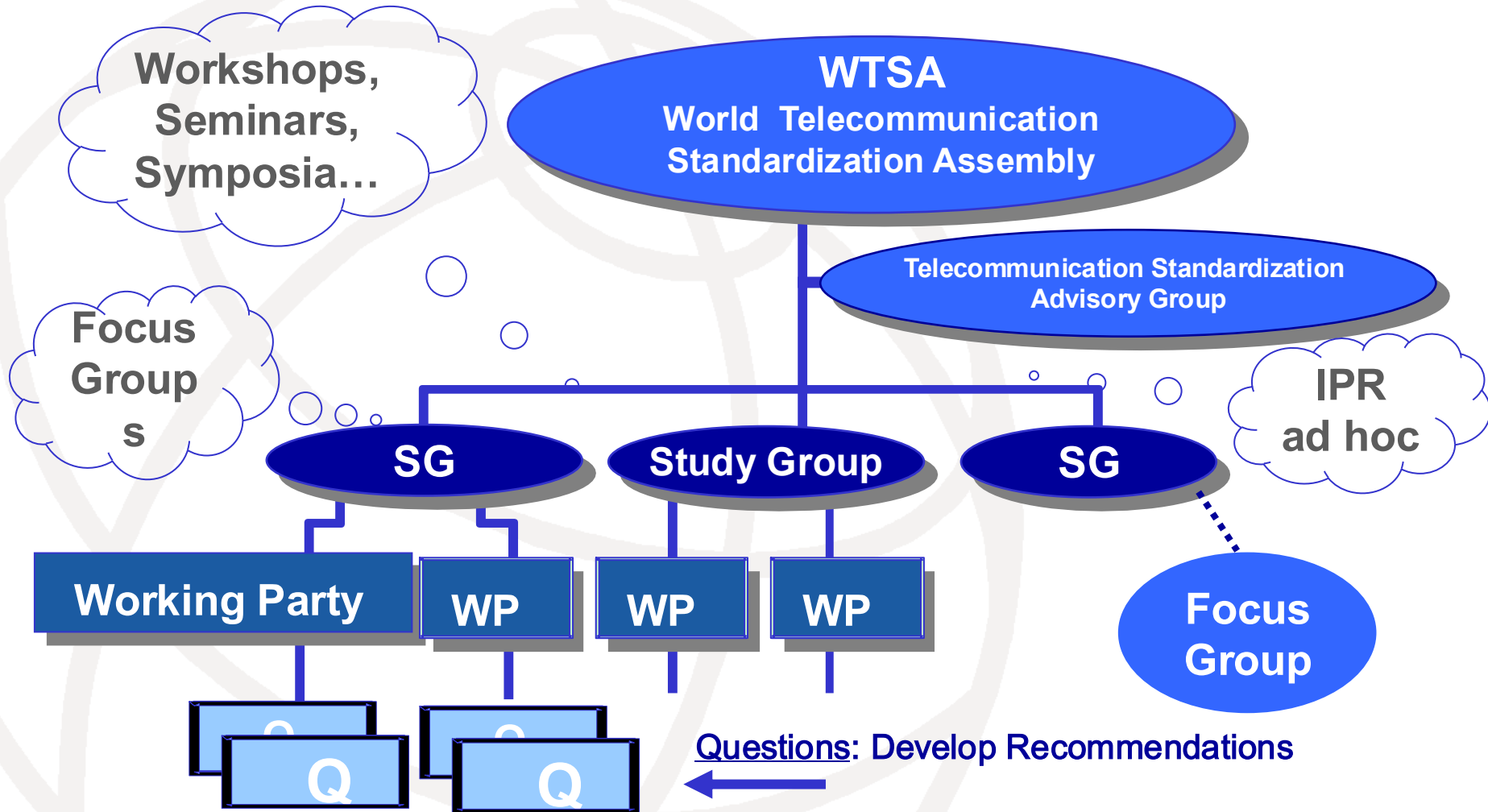
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ITU Structure



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ITU-T Structure



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ITU-T Objectives

- Develop and publish standards for global ICT interoperability
- Identify areas for future standardization
- Provide an attractive and effective forum for the development of international standards
- Promote the value of ITU standards
- Disseminate information and know-how
- Cooperate and collaborate
- Provide support and assistance



ITU-T Key Features

- Truly global public/private partnership
- 95% of work is done by private sector
- Continuously adapting to market needs
- Pre-eminent global ICT standards body



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ITU-T provides Broadband Access

- Copper:
 - Hundreds of millions use ITU-T's DSL
 - Up to 200Mbit/s aggregate with VDSL 2
- Cable:
 - IPCablecom
- Optical access:
 - ITU-T's GPON allows up to 2.5Gbit/s
 - New types of optical fibre for access networks



GPON interoperability pavilion
Nxtcomm, Chicago, 2007

ITU-T puts the Super in Information *Super* Highway

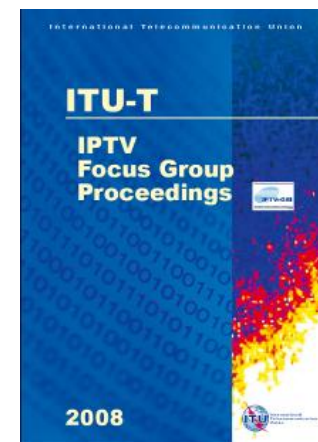
- Optical transport now to 100 Gbit/s
- Carrier class Ethernet
- Carrier class MPLS (MPLS-TP)
- Evolution towards an All Optical Networks (AON)



Next Generation Networks



- Telecoms revolution:
From circuits to packets
- Managed and secured
- With Quality of Service
- Saving money for customers and service providers
- IPTV standards well advanced



The network knocks at your door

- Home Networking to achieve interoperability on a global scale
 - Converged architecture and services
 - Next generation set-top box
 - PC World (US) 13.12.08: "The powerful world standards organization ...[ITU].. has reached agreement on G.hn a set of specifications that would encompass phone lines, power lines, and coaxial cable to provide HDTV room to room..."

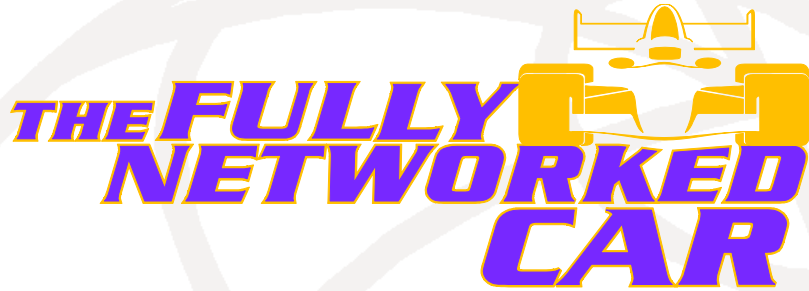


ITU-T's quantum leaps in speech, audio and video quality

- Emmy award received on behalf of ISO, IEC & ITU
- Call for technical contributions for H.265
- Extension of work on speech coding to wideband



Intelligent Transport Systems: new work, new members



- ITU, ISO and IEC and Geneva Motor Show
 - Annual Event
- New work:
 - Wideband communication in cars
 - Vehicle gateway protocol



Safety in the cyber world

- Identity management
- Security standards for:
 - NGN
 - IPTV
 - Home networks
 - Ubiquitous sensor networks
 - Mobiles
- Traceback
- Countering spam



ICTs and Climate Change



UN Secretary-General,
Ban Ki-moon: "ITU is one of the
very important stakeholders in the
area of climate change."

- Checklist to ensure new standards take climate change into account
- Methodology to describe and estimate present and future user [energy] consumption of ICTs over their entire life-cycle



Participants in
Focus Group ICT
and Climate
Change

Emergency Communications

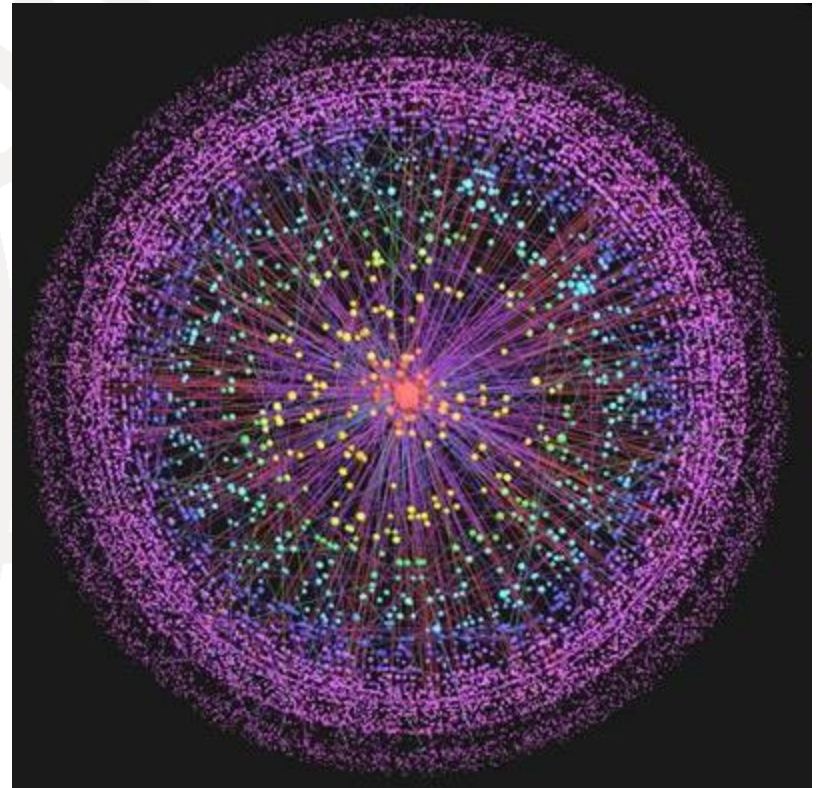
- Common Alerting Protocol (CAP)
 - A consistent method of delivery for warning messages
- Call priority schemes
 - Giving priority in disaster zones to emergency calls
- In Case of Emergency numbers



ITU has deployed satellite terminals to help restore communications in the aftermath of disasters around the world

Future networks

- Focus Group
- Collect and identify visions of future networks
- First meeting 6-10 July Geneva



Free Recommendations

- Since beginning of 2007, ITU-T Recommendations are available without charge.
- With only a small number of exceptions all in-force ITU-T Recommendations are available in PDF form via a simple mouse click:

itu.int/ITU-T/publications/recs.html



WTSA-08



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WTSA-08 Statistics

- Five regional preparatory meetings (Brazil, Ghana, Uzbekistan, Syria, Viet Nam) in association with Regional Development Forums on Bridging the Standards Gap in collaboration with BR, BDT and Regional Offices
- Resulted in regional common proposals from five regions
- WTSA adopted 21 new Resolutions, revised 27 existing Resolutions, adopted two new Recommendations and revised 7 existing Recommendations
- Total of 350 contributions
- 99 participating countries
- Over 1000 attendees
- 13 Ministers/Vice-Ministers
- VIPs spanning the world and the ICT industry sector



Key Resolutions

- **Resolution 44** Bridging the standardization gap between developing and developed countries
- **Resolution 58** to encourage the creation of national Computer Incident Response Teams (CIRTs) particularly for developing countries
- **Resolution 64** instructs ITU-T Study Groups 2 and 3 to study the allocation and economic aspects of IP addresses taking account of the ITU workshop on IPv6 in September 2008
- **Resolution 69** invites Members to refrain from taking any unilateral and/or discriminatory actions that could impede another Member State to access public Internet sites, within the spirit of Article 1 of the ITU Constitution and WSIS principles and to report any such incident to TSB.

Key Resolutions (2)

- **Resolution 70** encourages more work in the field of telecommunication/ICT accessibility for persons with disabilities
- **Resolution 71** to encourage cooperation between ITU-T and academia, universities and their associated research establishments, and invite Council to consider reduced fee
- **Resolution 73** on ICTs and Climate Change encourages the membership to work towards reductions in greenhouse gas in line with the United Nations Framework Convention on Climate Change
- **Resolution 74** instructs Director TSB to propose to Council reduced ITU-T fee for Sector Members from developing countries based on ITU-D model, and that it include its consideration of this matter in preparation of PP-10
- **Resolution 76** requires ITU-T to develop conformance and interoperability testing Recommendations as quickly as possible

Recommendations

- **Recommendation ITU-T D.50** asks that international Internet connection arrangements take into account the possible need for compensation for the value of elements such as traffic flow, number of routes, geographical coverage and cost of international transmission, and the possible application of network externalities.
- **Recommendation ITU-T D.156** asks that developing countries examine appropriateness of a network externality premium on incoming international traffic from the operators of developed networks to the operators of developing-country networks to fund extending networks in developing countries

Action Plan



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Industry Advisory Group

- Resolution 68 and GSS proposal:
 - High-level industry executives
 - Identify and coordinate priorities and subjects to minimize number of forums/consortia
 - Consult first with developing countries
 - Report to next WTSA



Council Group to be established

- **Resolution 75** requests that Council establish a group on Internet public policy issues to be integrated within the Council WG on WSIS
- Implemented by Council-08 and first meeting held February

Reports to Council-09

- **Resolution 76** Compatibility and Interoperability
 - Expert advisory group has been established to assist TSB to develop the Report to Council-09
 - Consultant appointed to assist TSB
 - Subject of later presentation

Reports to Council-09 (2)

- **Resolution 64** on allocation and economic aspects of IP addresses
 - Questionnaire will be issued soon to identify regional needs of developing countries in association with BDT
 - New web page on IPv6 soon
 - Organize seminars for developing countries on IPv6
 - TSB conduct study on IPv6 address allocation and registration for interested countries
 - Report to Council-2009

Reports to Council-09 (3)

- Director will propose to Council-09 that new members from developing countries can join ITU-T on level of financial contribution equal to that in ITU-D and report to PP-10 (**Resolution 74**)
- Director will invite the ITU Council to consider the admission of academic institutions, universities and their associated research establishments in the work of ITU-T as Sector Members or Associates, at a reduced level of financial contribution, particularly academic institutions of developing countries (**Resolution 71**)

Bridging the Standards Gap

Resolution 44 actions plan has 4 programmes:

- Programme 1: Strengthening standard-making capabilities
- Programme 2: Assisting developing countries in enhancing efforts in respect of standards application
- Programme 3: Human resource building
- Programme 4: Flagship groups for bridging the standardization gap
- Director establishing an implementation group within TSB which organizes, mobilizes resources, coordinates efforts and monitors work related to the action plan

Assistance to Developing Countries

- **Resolutions 17, 44, 56, 59, 72 and more:**
 - Organize workshops and seminars in the regions concerns (including related to human exposure to electromagnetic fields)
 - Support regional activities and study group VCs from developing countries
 - More meetings in regions
 - Regional and Flagship groups
 - Remote participation
 - Provide fellowships to all ITU-T Study Group and TSAG meetings

Actions

- Increased number of study group meetings in the regions planned
- New regional group meetings planned
- Workshops in regions on implementation of WTSA-08 actions and application of ITU-T Recommendations in regions
- Fellowships now available for all ITU-T study group meetings
- New ITU-T Flagship Group for Latin America
- Study Group 3 Regional Group for Latin America
Lima, Peru - 22 to 26 June 2009
- ICT and Climate Change Symposium, Quito, Ecuador 9 to 11 July 2009 preceded on 8 July by a briefing on the ITU implementation of the WTSA-08 decisions
- NGN-GSI and ITU-T SGs 11 and 13 - Mar del Plata, Argentina 2 to 12 September preceded by the "Kaleidoscope" 31 August and 1 September 2009

Conclusions

- ITU world's pre-eminent global ICT standards body
- Bridging the standardization gap recognised as essential to ITU's mission to Connect the World
- New team of chairmen and vice-chairmen from 33 countries
- First SG15 meeting since WTSA-08 was completely paperless, had largest ever participation (367 delegates) largest ever number of contributions (336) consented 28 Recommendations including new high profile global standard on home networking:
- Full list of WTSA Resolutions is at:
<http://www.itu.int/publ/T-RES/e>